



Requirement Specification

Shop Product Sale and Billing System



Objective

To develop a system that manages product details, stock details, sales, quantity management and billing for a shop, ensuring accurate stock management, proper calculation and easy record keeping.



Functional Requirements

1. Manage Product Details

- Store Product ID, Product Name, Category, Price and other basic product information.

2. Manage Stock Details

- Store available quantity of each product.
- Display current stock and stock status.
- Update stock automatically after a sale.

3. Manage Sale Details

- Record Sale ID, Product ID, Customer details,
- Sale Date, Quantity Sold, Sale Price and Total Amount.

4. Quantity Management

- Accept quantity required by the customer.
- Check whether the requested quantity is available.
- Prevent the sale when stock is insufficient.
- Reduce stock after a successful sale.

5. Bill Calculation

- Calculate product amount (Quantity × Price).
- Calculate subtotal.
- Apply discount and tax, if applicable.
- Calculate final bill amount.

6. Billing

- Generate a unique Bill ID.
- Display bill date, product details, quantity, price and total.
- Display final payable amount.



Non-Functional Requirements

1. The system should be easy to use.
2. Data should be accurate and consistent.
3. Product and sales information should be stored securely.
4. Billing calculations should be fast and reliable.
5. The system should prevent incorrect stock updates.
6. The database should support backup and recovery.



Inputs

- Product ID
- Quantity
- Sale information
- Discount / tax information (if applicable)



Outputs

- Product details
- Available stock
- Sale amount
- Updated stock
- Final bill



Basic System Operations

1. Add / update product details.
2. Check and manage stock.
3. Record sales and update stock.
4. Calculate bill amount.
5. Generate and print bill.
6. View sales and stock reports.



Algorithm

Shop Product Sale and Billing System



Algorithm

- 1 Start.
- 2 Enter or select the Product ID.
- 3 Retrieve and display the product details.
- 4 Display the available stock quantity.
- 5 Enter the quantity required by the customer.
- 6 Check whether the requested quantity is available.
- 7 If the quantity is not available, then
 - Display "Insufficient Stock".
 - Stop the transaction.

Else

- 8 Retrieve the product price.
- 9 Calculate product amount:
 - $\text{Amount} = \text{Quantity} \times \text{Price}$
- 10 Calculate the subtotal.
- 11 Apply discount and tax, if applicable.
- 12 Calculate the final bill amount.
- 13 Generate a unique Bill ID.
- 14 Update stock:

$$\text{New Stock} = \text{Available Stock} - \text{Quantity Sold}$$

- 15 Save the sale details.
- 16 Display the final bill.

- 17 Stop.



Key Formula

$$\text{Amount} = \text{Quantity} \times \text{Price}$$

Other Calculations:

- Subtotal = Σ Amount
- Discount = (Subtotal $\times$ Discount %)
- Tax = (Subtotal $\times$ Tax %)
- Final Bill Amount =
Subtotal – Discount + Tax



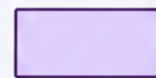
Symbols Used



Start / End



Input / Output



Process

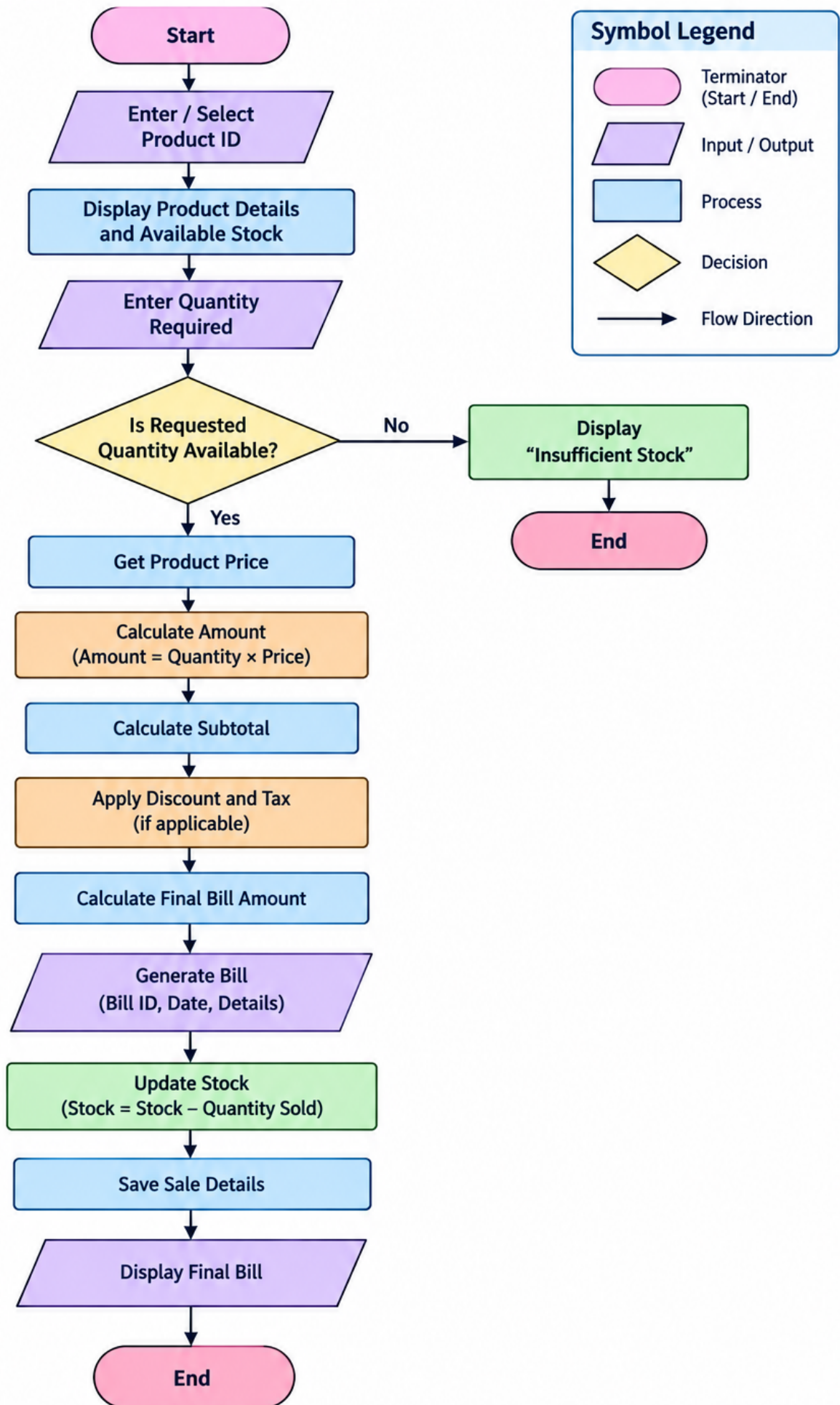


Decision

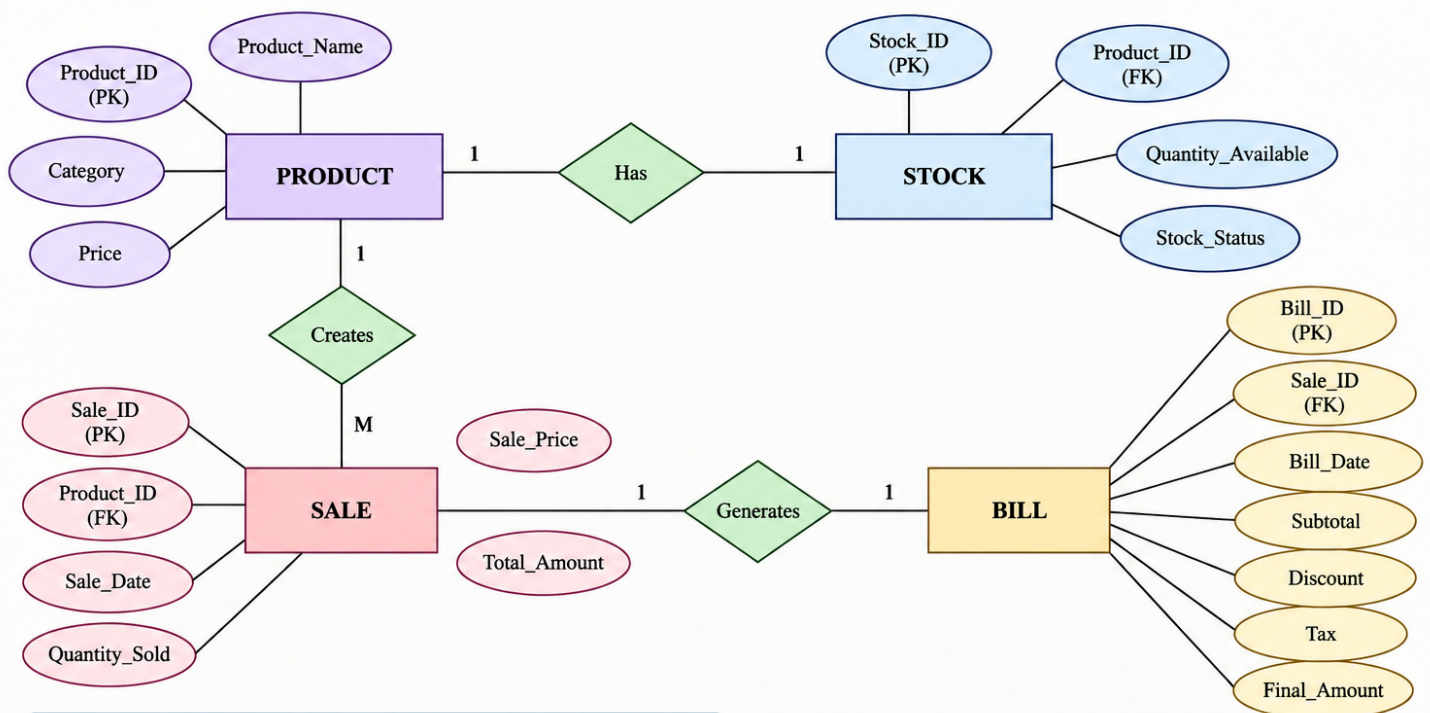


Flow Direction

Flowchart – Shop Product Sale and Billing System



ER Diagram – Shop Product Sale and Billing System



Legend:

= Entity
 = Relationship
 = Attribute

PK = Primary Key
 FK = Foreign Key
 1 = One
 M = Many

Relationship Summary:

1. **PRODUCT – STOCK** : 1 : 1 (One product has one stock record)
2. **PRODUCT – SALE** : 1 : M (One product can have many sales)
3. **SALE – BILL** : 1 : 1 (One sale generates one bill)